

Comparison of Embedded System Design for Industrial Applications

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Abstract—This paper presents a survey on embedded systems design and applications. Several platforms for embedded systems, including microcontrollers, microprocessors, field-programmable gate arrays, digital signal processors, and application-specific integrated circuits are discussed and compared. A survey of embedded system-based industrial applications is presented. Examples of real-life design decisions specific to development of such systems are also presented. The carefully selected three design case study examples include industrial control of wind tunnel with emphasis on actuator control, a mobile robot navigation system with emphasis on integration and synchronization of several subsystems, and optimized implementation of computationally intensive control system on a small microcontroller system.

Index Terms—Embedded systems.

I. INTRODUCTION

EMBEDDED systems can be found everywhere in daily life, from electrical commodities and appliances, to non-linear compensation mechanism, complex automation systems and adaptive control systems. Comparing with computer platform, embedded systems normally presents much less power of computation and very limited memory size. However, for solving particular real-time tasks, the disadvantages above turn around to be merits: embedded systems are less expensive and much easier to design. The simpler design is reflected in both hardware and software. Fixed design or only limited variations of hardware allow for use of simplified operating systems (OS) that allow for predictable, real-time operation, or even direct implementation of applications without any formal OS.

This paper is organized as follows. In Section II, four major platforms for embedded systems implementation, including microcontrollers and microprocessors, field-programmable gate arrays (FPGAs), digital signal processors, and application-specific integrated circuits are introduced in details. Section III presents the overview of the recently accepted papers on the subject of the various applications of embedded systems. Section IV gives several examples of applications as examples of embedded system design.

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II. DESCRIPTION OF EMBEDDED SYSTEM PLATFORMS

The embedded systems are normally defined as the software implemented in hardware in order to realize specified real-time functionalities. The normally used soft-core processing hardware includes microcontrollers, microprocessors, FPGAs, digital signal processors (DSPs), and application-specific integrated circuits (ASICs), each of which has its own properties.

A. Microcontrollers and Microprocessors

For many years only microcontrollers and microprocessors were applied as the only efficient way to implement embedded systems, because of their programmable functionalities. The hardware architecture of microprocessors is fixed and generic within a given subclass which makes the platform low cost. They are capable of executing sequences of basic instructions that are typically stored in persistent read-only memory (ROM) or more recently in on-chip FLASH memory.

Microcontrollers (MCUs) are typically manufactured with memory and some digital and analog peripherals integrated with a processor core on one chip. In order to reduce manufacturing cost and operating power, some microcontrollers are designed to use very short word length, such as four-bit words. They have very little random-access memory (RAM) and run clocked at a kilo Hertz range frequency. Also, those microcontrollers are able to retain partial functionality while the reminder of their circuit is suspended when waiting for an event or interrupt.

On the other end, microcontrollers may operate using 32- or even 64-bit words, be clocked at a hundred Mega Hertz range and have sufficient computational power to perform functionality of a DSP. In almost every case, however, internal ROM and minimal amount of RAM, some programmable interval timers, digital input-output circuitry, and some forms of serial communication interface are integrated into the chip while external memory bus might be left out from their design.

Microprocessors were developed as a single chip implementation of central processing units. Early embedded system utilized them. However, with development of more advanced and efficient manufacturing technology, the main applications of microprocessors remain in computing technology. In many embedded system applications, microprocessors are replaced by already discussed microcontrollers. Advanced microprocessor designs include several CPUs (multicore) on one chip, RAM memory cache due to latency caused by use of RAM external to the microprocessor chip, and some hardware support to implement virtual memory addressing.

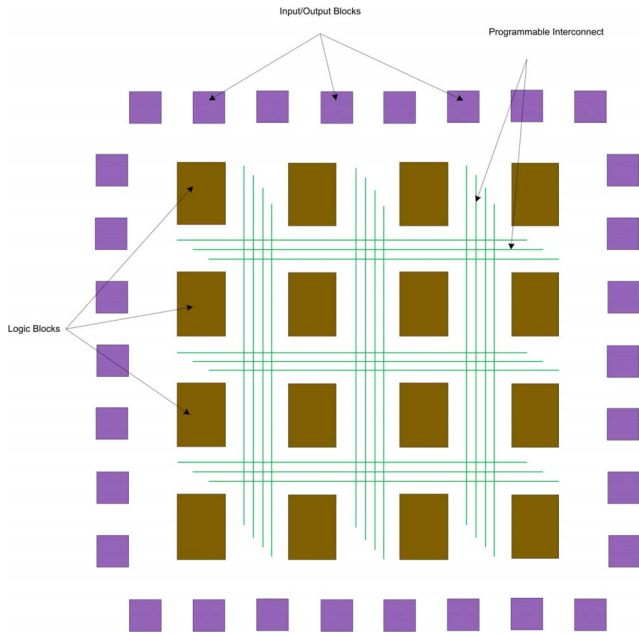


Fig. 1. Architecture of FPGAs.

B. Field-Programmable Gate Arrays (FPGAs)

FPGAs were developed for digital embedded system based on the idea of using arrays of custom logic blocks (LBs) surrounded by a perimeter of I/O blocks (IOBs), all of which could be assembled arbitrarily (Fig. 1).

FPGAs take the advantages of high operation speed, reconfiguration capability, very large number of components, and supported protocols. In embedded systems, FPGAs are used in two ways: either to implement the desired functionalities directly in the digital logic, or by implementing the architecture of a microprocessor—so called soft processor core, and desired microcontroller peripherals. The latter scenario became very popular in recent years as the FPGA prices reduced significantly, and could compete with microcontrollers. Use of FPGAs allows also for easy design of additional custom hardware accelerators that implement in hardware certain time consuming computations. Rodriguez-Andina *et al.* presented a thorough study of evolution of capabilities of FPGAs and design tools [1]. Monmasson *et al.* presented a very compact state-of-the-art tutorial demonstrating balancing use of hardware and software in design of a Kalman filter based AC driver controller [2]. In [3], Monmasson and Cirstea provided the overview of design techniques employed in design of various FPGA-based industrial controllers. Furthermore, capability to perform partial reprogramming on the fly of an FPGA leads to the practical implementation of an old idea of reconfigurable computing [4], [5].

FPGA technology is very valuable for one more reason. It makes possible to replace failed digital components in legacy systems [6]–[8]. Many industrial control systems were designed when expected lifetime of its components was about 25 years while presently the average design life cycle of used components is only about two years [9]. That makes it impossible to replace such failed components unless special steps are taken

such as stock piling of them. Another, more costly and time consuming choice is to redesign and replace whole subsystems.

C. Digital Signal Processors (DSPs)

Digital signal processors (DSPs) are designed to have embedded multipliers and DSP blocks, which allow complex arithmetic operations to be performed, so they are easy for high-level programming implementation. When compared with microcontrollers, one of the primary advantages of DSP is availability of a single cycle multiply and accumulation operation. Also, DSPs have parallel processing capabilities and integrated memory blocks, which largely enhanced the processing speed. Some DSP architectures, called digital signal controllers (DSC), are optimized for control applications and contain control-oriented peripherals such as PWM generators, watchdog timers and fast response interrupts. However, DSPs require much higher cost comparing with FPGAs. Usually, DSPs are applied for image and audio signal processing when use of microcontrollers is not possible due to computational limitations. Their primary application type in industry is motor controller.

D. Application-Specific Integrated Circuits

As another competitive type of implementation platform for embedded systems, application-specific integrated circuits (ASICs) have the advantages of high quality performance, low power consumption and low cost. In order to expedite the design process ASICs are built from composition of so called standard cells. Over the years, the design tools improved while maximum complexity and functionality increased. Current designs may include standard cells such as up to 32-bit processors, ROM, RAM, EEPROM, Flash and other large complex blocks. Already discussed FPGAs are frequently used for rapid prototyping [10] and replaced this technology for low quantity production. Use of ASICs is feasible only for manufacturing high quantity and long series due to higher initial engineering cost [10].

Besides the software-in-hardware embedded systems, the concept “embedded” is also extended as the merging of several technologies.

III. OVERVIEW OF EMBEDDED SYSTEM APPLICATIONS

This paper presents a survey on various applications of embedded systems in industrial fields. Recent publications on the subject in the IEEE TRANSACTIONS ON INDUSTRIAL ELECTRONICS are studied and classified into the following groups.

A. Nonlinear Compensation

Cotton *et al.* implant arbitrarily connected neural networks in microcontrollers used for nonlinear compensation [11].

B. Automation Systems

Ildirin *et al.* implement a software voting system on microprocessor with Safety Integrity Level 4 [12].

DouKas *et al.* present a brilliant research on the embedded framework to construct the paradigms of component and model-based engineering for industrial control and automation systems under a real-time Linux environment. A robotic arm is applied as an example to illustrate the functionality of the proposed framework and proposing experimental results are obtained [13].

C. Adaptive Control Systems

Špinka *et al.* introduce a reconfigurable control system used for small unmanned aerial vehicles design [14].

Monmasson and Cirstea propose to use FPGA-based controllers designed using various optimization techniques such as A³ and/or pipelining in order to achieve very low latency of execution of the control algorithm in the range of microseconds, unavailable in purely programmed solutions [3].

Onat *et al.* propose the Model Based Predictive Networked Control Systems (MBPNCS), with improved control stability by compensating random delays and data loss in communication networks [15].

Suetake *et al.* develop a compact embedded fuzzy inference system used for voltage-frequency speed control of induction motor [16].

Calabrese *et al.* propose an embedded multi-valued control algorithm used for ceramic manufacturing [17].

Silva *et al.* implement reconfigurable logic controllers (RLCs) using a novel matrix model to describe Petri nets (PNs) [18].

D. Image and Audio Processing

Belbachir *et al.* develop a high-speed embedded vision recognition and classification system, using a neuromorphic dual-line vision sensor and signal processing technology [19].

Weber *et al.* develop a configurable system to perform frequency-diverse target detection used for real-time ultrasonic imaging [20].

Cheng *et al.* develop an automatic speech recognition system using hardware-software co-processing [21].

E. Internet Services

Pérez *et al.* introduce a web service on chip (WSoc) system to run a particular web service in an application-specific integrated circuit, purposed to implement more cost effective and zero-management service-oriented architecture network devices [22].

Sziebig *et al.* develop a multimedia educational system used for distant learning [23].

F. Communication Systems

Baronti *et al.* present the design and verification of hardware building blocks, including a FlexRay transceiver and a SpaceWire router with related interface, in high-speed and fault-tolerant in-vehicle networks [24].

Guo *et al.* introduce a wireless sensor network with optimized localization using Gauss-Newton algorithm and Particle Swarm Optimization [25].

Barranco *et al.* develop an innovative CAN-compliant star topology with enhanced fault-treatment mechanisms [26].

Smart embedded cyber sensors are essential components of resilient and secure control systems of critical infrastructures, such as Supervisory Control and Data Acquisition (SCADA), nuclear plant, and smart grid systems. These embedded systems frequently employ learning algorithms for anomaly based network security cyber sensors together with its hardware implementation. The learning algorithms are here specifically developed to comply with the constrained computational requirements of low-cost embedded network security cyber sensors [27], [28].

G. System Diagnosis and Noise/Fault Analysis

Based on microcontroller, Kim *et al.* design a smoothing predictive redundancy system and evaluate the fault-tolerant ability based on its microcontroller implementation [29].

Pérez *et al.* develop a novel network intrusion detection system embedded in a smart sensor inspired device, purposed to reduce the huge volume of management tasks [30].

Miranda *et al.* discuss the applicability of implementing components in embedded systems using FPGAs, in order to cope with the obsolescence problems of microprocessors [31].

Li *et al.* implement the signal hardware-in-the-loop (HIL) simulation to estimate the performance of a wind turbine generator which is coupled with a hybrid energy storage system [32].

Ordóñez *et al.* develop an embedded frequency response analyzer in order to monitor and measure the different electrochemical processes occurred inside the fuel cells (FCs) [33].

H. Robotic Platforms

Various applications of embedded systems supporting mobile robot teleoperation can be found in industrial automation, freight handling or transportation, nuclear waste manipulation, or explosives disposal. Some applications tackle the neural network haptic teleoperation of mobile robots, such as Self-Organizing Fuzzy Adaptive Mapping [34]. Such embedded system provides the operator with an improved depth-judgment and increased obstacle awareness.

Recently, multiple-robot systems with embedded decentralized control mechanisms have been intensively researched. The examples range from single-operator manual control of multi-robot system [35], where each mobile robot maintains an embedded controller with built-in swarm behavior that controls low level tasks such as formation keeping and obstacle avoidance. Such multirobot systems can also be used for multiple targets optimization for guiding robots in high risk environments such as chemical spills or radioactive environments [36].

Embedded systems are frequently used in intelligent transportation systems (ITS), surveillance, scheduling, planning or industrial automation. Examples include spatio-temporal sampling from embedded wireless position sensors, such as in [37], to achieve the online recognition and extraction of the set of the most significant places and provide dynamic online risk assessment. Such transportation applications also entail embedded sensors for object tracking, lately expanding the idea from GPS coordinates to wireless network triangulation [38].

Robotics applications frequently deploy computational intelligence based embedded systems for control of actuators whether robotic hands or more sophisticated parallel robotic

structures are in case [39]–[41]. Here, various tasks ranging from data collection, communication with user interface, control of the actuator are handled by embedded systems based on fuzzy, neuro or evolutionary technologies.

I. Circuit Design

Huang *et al.* discuss a programmable system-on-chip (SoC) design methodology to integrate multiple heterogeneous SoC design projects in a single chip, purposed to reduce the total silicon prototyping cost [42].

Muñoz *et al.* implement an adaptive filter with robust ability to impulsive noise, in both hardware description language and a high-level synthesis design tool [43].

Oliveira *et al.* propose a time efficient Simultaneous MultiThreading (SMT) processor embedded with Advanced Real-Time Processor Architecture–MultiThreaded (ARPA-MT) [44].

J. Education Platforms

Baese *et al.*, provide an intimate knowledge of the relationship between the microprocessors design and the development tools with two teaching modules based on ADL. The URUSC model and the Educational RISC process model are developed [45].

Cardoso presents an approach to teach the design of non-programmable application-specific architectures using VHDL, logic and physical synthesis tools and FPGAs. The approach replies on mini-projects that resemble typical problems that students may face in real-life concerning the design of application-specific architectures [46].

Hercog *et al.* present a rapid control prototyping (RCP) system, based on commercially available software and custom in-house developed hardware. This RCP system successfully combines the well-known simulation program MATLAB and the custom DSP-based floating point motor controller, which is suitable for educational processes as well as motor control research [47].

Ibrahim describes the design of a digital filter using a low-cost microcontroller as the processing element. It aims to teach students the basic hardware and software implementation of digital filters. Both FIR and IIR type filters can easily be implemented with the system [48].

Farias *et al.*, introduce a novel approach to building virtual laboratories of embedded control systems using TrueTime and Easy Java Simulations. The combination of these two tools conforms a powerful, yet simple, approach to the creation of effective pedagogic simulation of real-time control systems [49].

IV. CASE STUDY EXAMPLES OF EMBEDDED SYSTEMS DESIGN

The following three case studies illustrate sample use of microcontrollers as components of larger industrial systems. The case study is limited to microcontrollers as sample use of FPGAs has been covered recently elsewhere [1], [3]. The selected case studies illustrate use of microcontrollers both as autonomous components (Section IV-A), and as parts of a larger system with components that use advanced communication (Section IV-B). Finally, a detailed example of implementation of a control algorithm on a platform with limited resources is presented in Section IV-C.



Fig. 2. Bradley University Subsonic Wind Tunnel.

A. Local Feedback Controller

The Bradley University Subsonic Wind Tunnel, shown in Fig. 2, is an open-loop induction tunnel with a test section that is 28 cm high, 36 cm wide, and 60 cm long. Conditioned air is drawn from the surrounding room through the test section by a 50 hp rotary fan. The fan exhausts the air out of the room helping to maintain constant operating conditions in the test section. The wind speed in the section is regulated using dampers located immediately upstream of the fan. Because the wind tunnel is an induction, open-loop type, the turbulent intensity in the test section is relatively low and has been measured at about 0.5%. The velocity boundary layer is approximately 1 cm thick at the exit of the test section leaving a large core of undisturbed flow. Two actuators are mounted inside the tunnel. One allows for changing the angle of the tested object. Another allows changing the position of the wind speed probe behind the tested object. Additional wind speed probe is mounted upstream of the object in order to measure the wind speed.

The tunnel can be controlled either manually using push buttons, or from a computer using LabView and GPIB interface. While it is possible to implement the complete actuator control algorithm in LabView, use of a dedicated simple embedded microcontroller system allows moving the implementation of the real-time control algorithm to the firmware of the Wind Tunnel. The attached computer performance is no longer critical. The LabView program on the computer implements automated data acquisition using multiple measurement points for varying object and wind probe positions and for different wind speeds.

Two schemes for interfacing the embedded system were initially considered. One would allow connecting LabView to the embedded system using USB-RS232 adapter, and UART interface, as shown in Fig. 3(a). The second requires use of LabView-compatible data acquisition card as shown in Fig. 3(b). The first approach requires a microcontroller with more AD channels potentially both faster and with higher data resolution, that allows to eliminate the need for data acquisition card. Since such microcontrollers are significantly less expensive than dedicated data acquisition cards, this approach should be favored if mass production of such wind tunnels was considered.

The latter approach makes two subsystems more isolated from each other which is one of the favorable solutions because computers and installed software are becoming obsolete quicker and are replaced more frequently than the wind tunnel that they control. Historically, data acquisition card vendors have been reliable to provide updated drivers for their hardware as computer equipment is updated. However, it requires a data

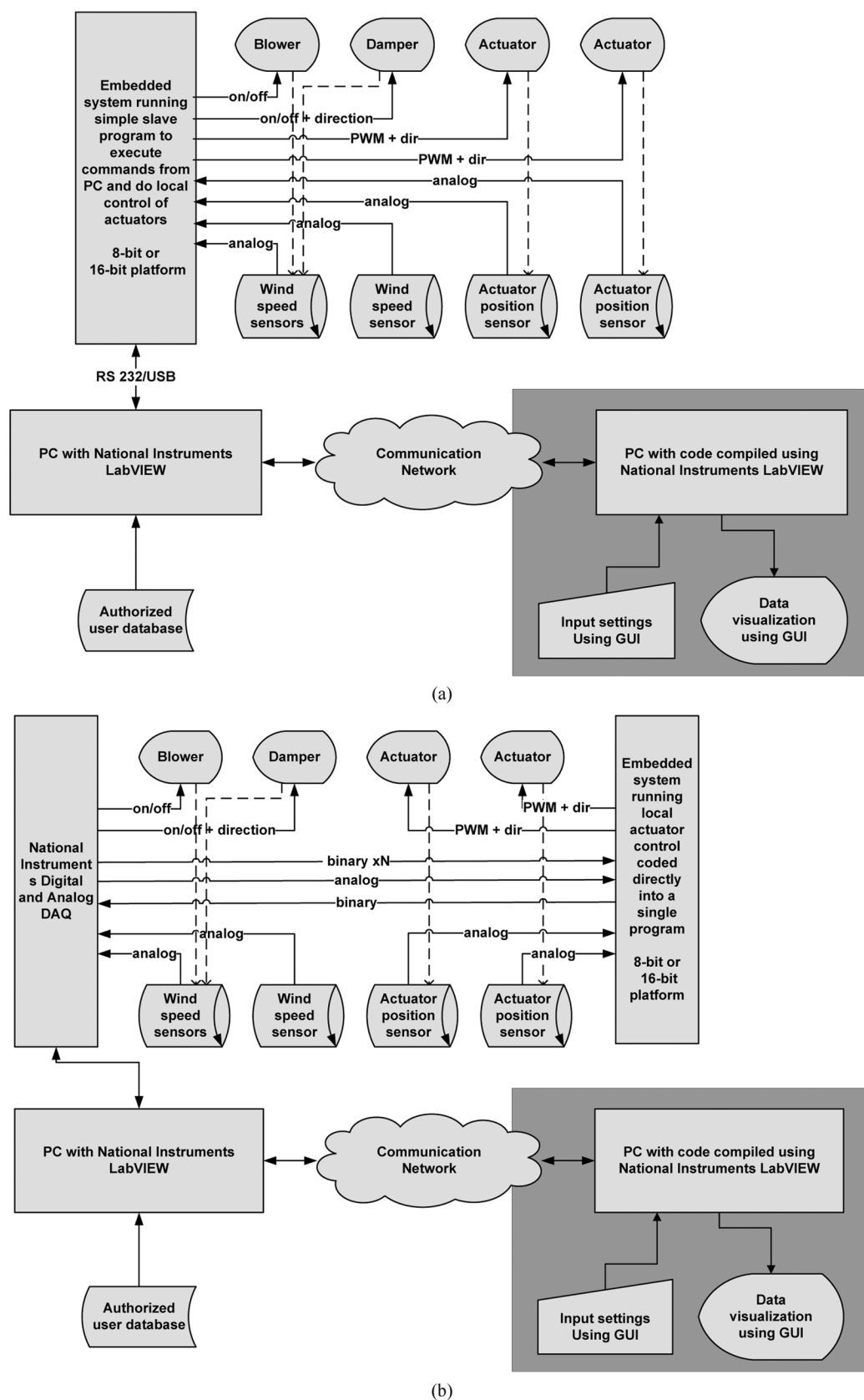


Fig. 3. Use of a microcontroller in Subsonic Wind Tunnel: (a) With a microcontroller performing all data acquisition. (b) With a dedicated data acquisition card and a microcontroller performing local control of selected actuators only.

acquisition card with more output channels which increases the manufacturing cost dramatically. Both block diagrams show the interaction between the microcontroller and other system components.

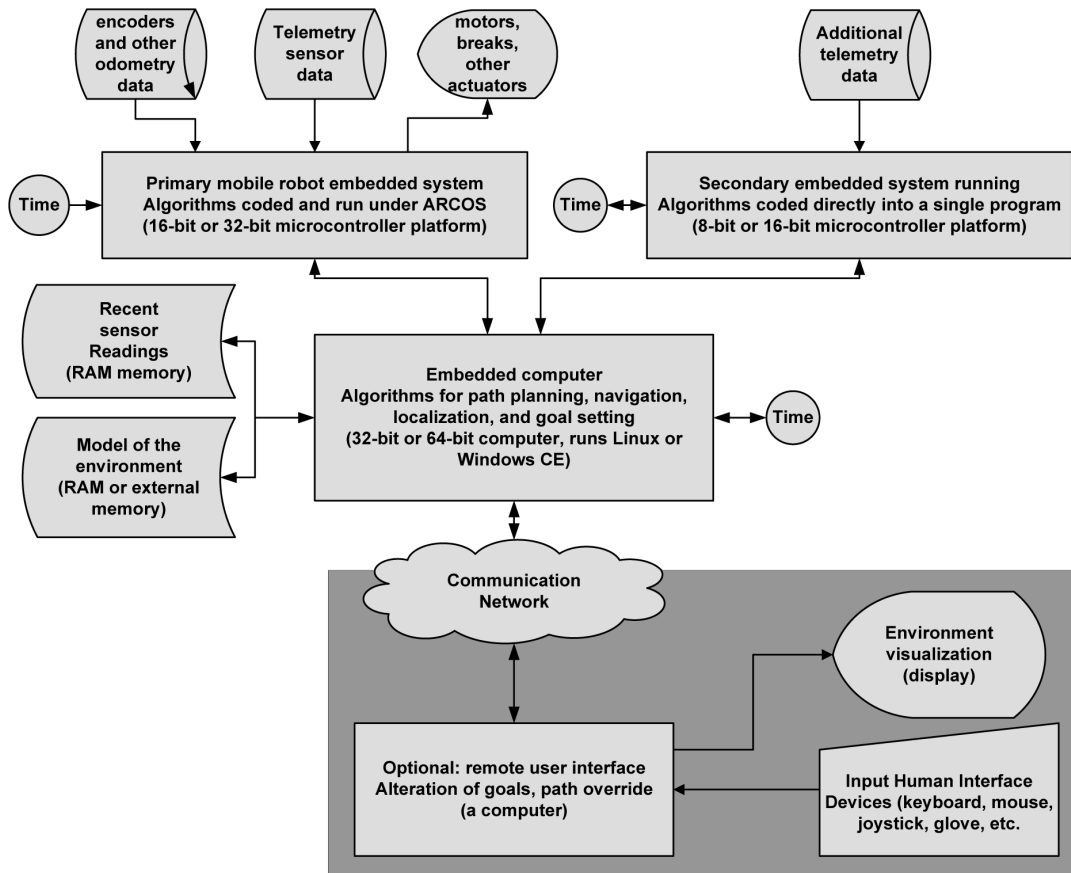


Fig. 4. Microcontrollers and embedded computer in Pioneer3-DX mobile robot navigation application.

The third possible approach, not considered here, requires the local actuator control loop to be implemented on the computer. That would eliminate the need for the embedded system, would still require more pricey data acquisition card, and furthermore, would put demand on the control program to run in hard real-time.

B. Sensor Interface for Mobile Robotic Platform

Pioneer3-DX is a mobile robotic platform developed by Adept Mobile Robots that belong to research robots category. Its base software architecture consists of at least two network-connected components. The first one is a differential drive based robotic platform with two motor-driven wheels with encoders, a set of distance sensors, and an embedded system that runs Advanced Robot Controller Operating System (ARCOS) responsible for controlling the robot movement and collecting data from sensors. The second component is an embedded computer platform that runs user-developed navigation software based on ARIA open source Pioneer SDK [50]. These two components can be connected either using either RS232 or IEEE 802.3 network. In this case study application, a third embedded has been introduced in order to interface additional telemetry sensors.

Fig. 4 shows the block diagram of the entire system. Controlling motors, performing dead reckoning, and collecting basic obstacle avoidance telemetry requires hard real-time operation. Making decision regarding current movement is a

soft real-time task. On the other hand, algorithms involved in deciding long term robot behavior and performing navigation calculations can be both complex and may require a lot of computational and memory resources. Although it is possible to implement both the mobile platform controller and the navigation software using one computing system such design would be unnecessarily complex. A microcontroller, or a group of microcontrollers can be used to perform the real-time tasks, and an embedded computer can be used to perform the navigation and to interface to additional resource consuming sensors such as cameras. The soft real-time task of computing robot movement is closely associated with long term navigation and thus should be implemented on the computer. Although it is not the case of Pioneer3-DX and ARIA, performing safety checks such as slowing or stopping the platform in proximity of obstacles may be more reliable if implemented on the microcontroller.

Optional remote computer implements control user interface in case the mobile system is not set up to be completely autonomous. Although direct remote control of the mobile platform is not possible because only soft real-time, computers were used, it is possible to set sequential goals to be achieved by the otherwise autonomous system which allows successful teleoperation even in case of significant communication latency between the mobile platform and the remote computer. Otherwise, an adequate communication channel would be required for direct remote control [51].

There is also an issue of communication latency among the local subsystems on the mobile platform. Even though

the microcontroller may have information collected from sensors ready to be communicated with little delay, it will enter the communication interface queue and be sent with certain delay. Furthermore, the operating system on the embedded computer system will most likely store the received data in a buffer, and the data will be processed with additional delay due to multitasking. In order to synchronize events again, each collected piece of information should contain a time stamp. Those time stamps are generated by the microcontroller at the time when data is collected using values read from hardware timer counter register immediately after a sensor reading is obtained. This allows for matching wheel odometry readings with telemetry readings, and to use a model of the mobile platform to interpolate its position for the events with time stamps that cannot be matched directly to odometry events. This process of reassembling of the robot state on the embedded computer is called *reflections* by the creators of ARIA.

The system used in this case study contains additional secondary microcontroller that collects data from additional sensors. In order to allow use of the same event matching technique by the secondary microcontroller its timer that is used for time stamping needs to be synchronized with the primary microcontroller that runs ARCOS. However, because of the already decided architecture of ARCOS on Pioneer3 the only obvious way to synchronize the secondary microcontroller is through the embedded computer that periodically sends the interpolated reflection of the primary microcontroller timer to the secondary microcontroller. Otherwise, existing techniques for synchronization of multiple networked embedded systems could be used, for example, [52].

C. Neural Network Implementation

Being capable of simulating any nonlinear relationship between stimulus and responses, neural networks have broad applications in various industrial fields, such as nonlinear compensation [53], [54], adaptive control [55], [56], image processing [42] and system diagnosis [43]. There are various PC platform based tools for neural networks design and training, such as the famous MATLAB Neural Network Toolbox (MNNT) and the powerful neural network trainer (NNT) [59]–[64], which is capable to handle arbitrarily connected neural network architectures with both first and second-order learning algorithms. Besides PC platform, neural networks are also implanted to embedded systems, such as FPGAs, DSPs and ARMs [65]–[71].

1) *Why Neural Networks:* Both neural networks and fuzzy inference systems are often considerable technologies for building nonlinear controllers. In order to illustrate the properties of each method, let us two design function approximators as the example.

The experiment of function approximation was carried on in the following scheme. The 25 points in Fig. 5(a) are supposed to be known and applied to design the approximators. Then, the 625 points in the same range [Fig. 5(b)] are applied to test the designed approximators. All the points can be obtained by the (1)

$$z = 4 \exp(-0.15(x-4)^2 - 0.5(y-3)^2) + 10^{-9}. \quad (1)$$

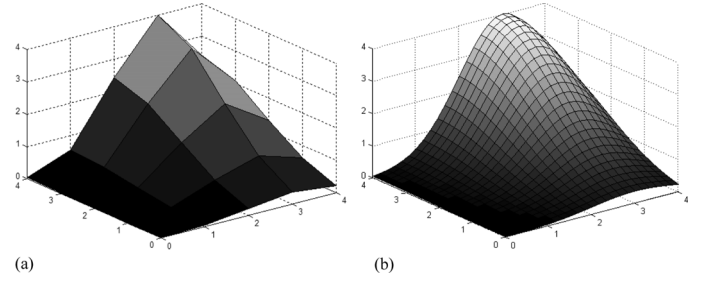


Fig. 5. Surface approximation problem. (a) $5 \times 5 = 25$ points for design. (b) $25 \times 25 = 625$ points for test.

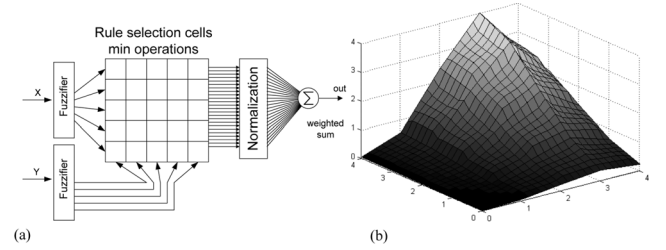


Fig. 6. Fuzzy inference system for the surface approximation problem. (a) TSK architecture with five membership functions in each direction. (b) Result surface, MSE = 0.0213.

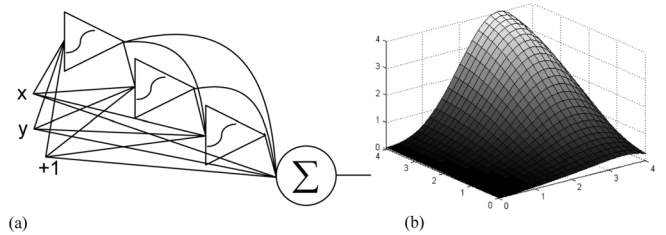


Fig. 7. Neural network for the surface approximation problem. (a) Four neurons in fully connected cascade network. (b) Result surface, MSE = 0.0015.

For fuzzy inference systems, the fuzzy table can be constructed using the 25 known points. Then, ten membership functions (five in each direction) are required. Fig. 6(a) shows the designed TSK fuzzy inference architecture and Fig. 6(b) shows the testing results.

For neural networks, the 25 known points can be applied as input patterns directly for training. The tried smallest network architecture is 4 neurons in fully connected cascade network, as shown in Fig. 7(a), and the related testing result is shown in Fig. 7(b).

Based on the experiment, the comparison of two different approximators is concluded in Table I. One may notice that neural networks have the advantages of small size, short code, fast computation, and accurate approximation, over fuzzy inference systems. However, before testing, neural networks require very complex training process which is not necessary for fuzzy inference systems.

2) *Computations in Microcontroller:* In this example, an inexpensive 8-bit microcontroller is used to build the trained neural networks. The implementation of neural networks in microprocessors is quite different from that on PC platform, because the computational power of microprocessors is much

TABLE I
COMPARISON OF NEURAL NETWORKS AND FUZZY INFERENCE SYSTEMS BASED
ON THE FUNCTION APPROXIMATION PROBLEM

Properties	Fuzzy systems	Neural networks
Design Complexity	No Training	Need Training
Number of Units	10	4
Code Length	1502	705
Processing Time (ms)	56.7790	3.4160
Error MSE	0.0213	0.0015

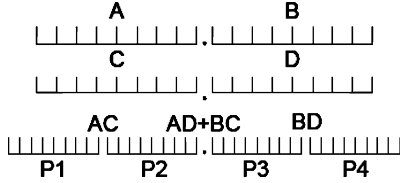


Fig. 8. Multiplication routine.

less than that with PC. The main problems for neural network implementation based on the microcontroller are: 1) floating data storage; 2) multiplication; and 3) hyperbolic tangent function approximation.

a) Floating Data Storage: In the assembly programming, there is no defined type for floating data. However, weight parameters for neural networks are almost floating data and require very precise values. In order to store the weights in microcontrollers with high accuracy, the example used a pseudo floating point method. First of all, weight values are scaled in order to avoid the round-off error and overflow condition. Then, the scaled weight values (without decimal point) are stored and extra memory is spent to remember the position of decimal points for related weight values. The pseudo floating point method is helpful for precise floating data storage, and it also improves the accuracy of floating data multiplications.

b) Multiplication: The hardware multiplier normally cannot handle floating point values and negative values. In this example, the multiplication routine, as shown in Fig. 8, is passed two 16-bit numbers (A.B and C.D), consisting of an 8-bit integer (A and C) and an 8-bit fraction portion (B and D). The multiplication results are 32-bit data.

Equation (2) shows the implementation of 16-bit multiplication routine (Fig. 8) using 8-bit multiplier. The result of AC is stored in P1 and P2; the result of BD is stored in P3 and P4; the sum of AD and BC is added to P2 and P3. This routine is very simple and does not require any shift or division operations

$$\text{res} = \frac{AC \times 256^2 + 256 \times (AD + BC) + BD}{256^2}. \quad (2)$$

c) Hyperbolic Tangent Function Approximation: Normally, hyperbolic tangent function (3) is adopted as the activation function in neural networks. Unlike high level languages (like C/C++), there is no exponential function in assembly language. So, it has to be approximated or replaced

$$y = \tanh x = \frac{2}{1 + \exp(-2x)} - 1. \quad (3)$$

Instead of using hyperbolic tangent function, the Elliott function (4) can be applied for neural network training. The Elliott function has the advantage of simplified computation and

TABLE II
APPROXIMATION RESULTS OF DIFFERENT POLYNOMIAL ORDERS

Approximation accuracy	1 st order	2 nd order	3 rd order
Stored Table Size	5	9	13
Maximum Error	0.0817	0.0045	0.00085

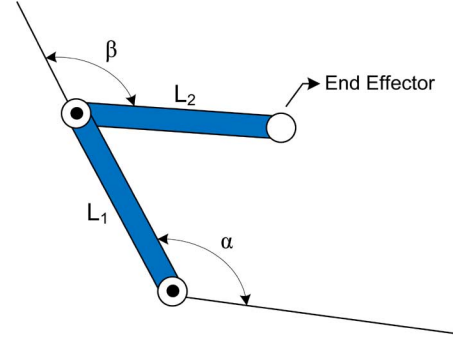


Fig. 9. Two-link planar manipulator.

it is proper for microcontroller implementation. However, in cases of nonlinear approximation with neural networks, Elliott function usually requires more neurons than hyperbolic tangent function [1]

$$y = \frac{x}{1 + |x|}. \quad (4)$$

Considering the microcontrollers without hardware dividers, the advantage of using Elliott activation function is gone and hyperbolic tangent function needs to be approximated by other methods, such as lookup-table (LUT) and polynomial approximation. The LUT method can perform very accurate approximation for given points, with the tradeoff of much memory for table storage. For polynomial approximation, high-order polynomials are required in order to achieve acceptable accuracy because of the nonlinearity of hyperbolic tangent function, so it is computational expensive.

In this example, the LUT method and polynomial approximation are blended, in order to obtain acceptable precision with limited table size, as presented in Table II.

3) Forward Kinematics Application: The neural network embedded microcontroller was applied to design a two-link planar manipulator (Fig. 9) which is purposed to determine the position and orientation of robot's end effectors when the joint angles change.

As shown in Fig. 8, the position coordinates of the two-link planar manipulator can be calculated by

$$x = L_1 \cos \alpha + L_2 \cos(\alpha + \beta) \quad (5)$$

$$y = L_1 \sin \alpha + L_2 \sin(\alpha + \beta). \quad (6)$$

With the data generated by (5) and (6), the neural network was firstly trained by the neural network trainer [44], [45] and then implemented in microcontroller. Through the serial port, the test data was transmitted to the microcontroller, and then the computation results obtained from the embedded neural network were returned to PC.

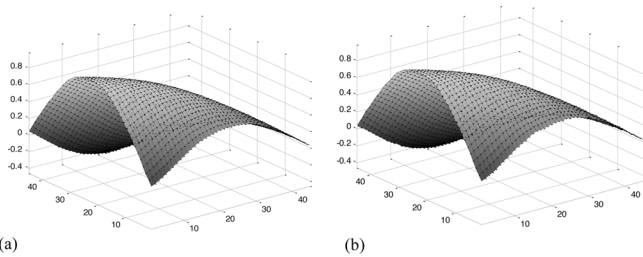


Fig. 10. Position x of the two-link planar manipulator. (a) Neural network in PC. (b) Neural network in microcontroller.

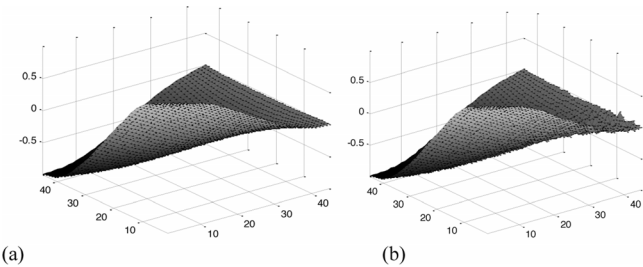


Fig. 11. Position y of the two-link planar manipulator. (a) Neural network in PC. (b) Neural network in microcontroller.

Figs. 10 and 11 present the testing results of the trained neural networks, based on PC and the microcontroller platforms separately.

V. CONCLUSION AND PERSPECTIVES

This paper introduced the current development of embedded systems. Various implementation platforms, including microcontrollers, microprocessors, FPGAs, DSPs, and ASICs, are discussed and compared. Three real applications of embedded systems are presented in order to illustrate issues that need to be considered and problems that need to be solved when designing embedded systems.

As sufficient work presented in the literature research above, it can be concluded that embedded systems have already maturely developed and can be designed to solve various very complex problems in industrial applications.

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